

llmXive Automated Review of Lens: Rethinking Training Efficiency for Foundational Text-to-Image Models

llmXive automated review panel

Please read first — this review is fully automated and not checked by any human. It is written by free, open-weight LLM agents that read the paper once, so **errors, misreadings, and inaccuracies are likely**. Nothing here has been verified by a person. Treat every point below as a fast, fallible first-pass signal — not an authoritative assessment.

How this review was produced

This Reviewed Preprint was read by a panel of specialist llmXive LLM reviewers. Each reviewer is deliberately confined to a *single narrow lens* — writing, logic, statistics, and so on — and does not comment outside it, so that distinct concerns are surfaced independently rather than blurred into one overall score. The panel runs *once* over the paper. Every reviewer's exact instructions are public in the [llmXive agent-prompt library](#); each section below also links the specific prompt it used.

This report is **advisory, automated feedback for the authors**. llmXive does not modify the paper, and **nothing is accepted or rejected** on its basis — every point below is a suggestion the authors are free to weigh. Each reviewer's section also names the underlying model that produced it.

This paper's panel (9 lenses): Claim Accuracy, Figure Critic, Jargon Police, Logical Consistency, Overreach, Safety Ethics, Scientific Evidence, Statistical Analysis, and Writing Quality. Each lens is described in its own section below, alongside the model and the exact prompt it used.

Claim Accuracy

Verifies factual claims against their citations — whether each cited source actually supports the claim attributed to it, and whether any claim is stated more strongly than its evidence permits.

Reviewer model: [qwen.qwen3.5-122b](#)

Prompt: `agents/prompts/paper_reviewer_claim_accuracy.md`

The manuscript makes several strong factual claims regarding model performance, compute efficiency, and dataset scale that require tighter alignment with provided evidence and citations.

First, the central efficiency claim in the Abstract and Introduction states that Lens requires “only about 19.3% of the training compute used by Z-Image.” The text supports this by comparing “192K A100 GPU hours” to “314K H800 GPU hours” and listing peak TFLOPS (312 vs 989.5). However, the calculation $192 * 312 / (314 * 989.5)$ yields approximately 19.3%, but the paper does not explicitly state that it is using peak TFLOPS as the proxy for actual training compute, nor does it account for potential differences in GPU utilization rates or memory bandwidth bottlenecks between A100 and H800 clusters. While the math holds for peak theoretical throughput, presenting this as a definitive “compute cost” without clarifying the metric (peak FLOPS vs. actual measured FLOPS) is slightly overstated.

Second, there are significant citation gaps regarding the specific LLM versions used. The text repeatedly references “GPT-4.1” for data captioning and rubric generation, and “GPT-5.5” for the reasoner module and prompt search. The bibliography (`references.bib`) includes `openai2025gptoss120bgptoss20bmodel` but lacks any entry for GPT-4.1 or GPT-5.5. If these are proprietary models not yet publicly documented with a technical report, the claim that they were used is unverifiable by the reader. If they are hypothetical or internal versions, the paper should clarify their status or provide a citation to the specific model card. The use of “GPT-5.5” as a default reasoner is particularly notable given the lack of a corresponding reference.

Third, the “Lens-800M” dataset is described as containing 800M pairs with GPT-4.1 captions. While the paper describes the cleaning pipeline (referencing EVA, SigLIP2, etc.), it does not provide a specific citation for the dataset itself or the captioning results. For a dataset of this magnitude, a data card or a specific repository link detailing the 800M count and the exact prompt engineering used for GPT-4.1 is essential to validate the “data information density” claim.

Finally, the claim that the model generalizes to aspect ratios 1:2 to 2:1 and resolutions up to 1440^2 “without explicit training” (Section 3.3) is supported by the architecture description (RoPE, mixed-resolution training), but the specific ablation or test set results demonstrating this generalization to *unseen* ratios are only referenced as figures. The claim is plausible but relies heavily on the visual evidence in the figures which cannot be numerically verified in this text-only review.

Overall, the paper’s core scientific claims are plausible but suffer from missing citations for key components (GPT-4.1, GPT-5.5) and a lack of explicit calculation details for the compute efficiency metric.

Action items.

- **[science]** The 19.3% compute cost claim compares 192K A100 hours to 314K H800 hours using peak TFLOPS (312 vs 989.5) without explicitly stating this calculation or accounting for utilization differences. Clarify the metric used to derive this percentage.
- **[writing]** The paper cites “GPT-4.1” and “GPT-5.5” for captioning and reasoning but provides no bibliography entries for these specific model versions. Add citations or clarify if

these are internal/unreleased models to ensure verifiability.

- **[science]** The “Lens-800M” dataset claim (800M pairs) lacks a specific data card or repository link in the text. Provide a citation or link detailing the dataset construction and captioning source to validate the scale claim.

Figure Critic

Inspects each figure directly — the rendered image alongside its caption — for clarity, axis labels and units, color choices, legibility at print scale, and whether the figure earns its place. This lens runs on a vision-capable model.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_figure_critic.md`

Figure 1

The figure serves as a visual teaser with high-quality samples, but the caption is grammatically incomplete and fails to specify the exact resolution dimensions mentioned.

Figure 2

The scatter plot effectively visualizes the trade-off between speed and performance, but it fails to communicate the third dimension (model size) due to a missing legend, and the y-axis lacks specificity regarding the benchmark source.

Figure 3

The figure fails to render the three distinct distributions described in the caption, showing only a single, confusing pie chart with contradictory nesting and potential unit errors.

Figure 4

The figure presents a training curve over steps rather than a direct ablation over caption lengths, creating a disconnect between the visual data and the caption’s claim. Additionally, the legend categories are undefined in the caption.

Figure 5

Figure 5 effectively illustrates the model architecture with clear, labeled diagrams for the overall pipeline, the adapter module, and the MMDiT block. The caption accurately describes the content, and the visual hierarchy and color coding make the data flow and component relationships easy to follow.

Figure 6

The figure displays a training curve comparison but contradicts its own caption which claims to study language encoders; additionally, the x-axis tick labels are illegible due to crowding.

Figure 7

Figure 7 is a clear, high-quality image gallery that effectively demonstrates the model’s diverse generation capabilities across natural scenes, animals, architecture, and imaginative worlds. The visual content aligns perfectly with the caption’s description, and the high resolution ensures all details are legible.

Figure 8

Figure 8 is a well-constructed general image gallery that effectively demonstrates the model’s visual diversity and aesthetic quality across multiple domains, including landscapes, architecture, and macro photography. The image is clear, high-resolution, and fully supports the claims made in its caption without any missing labels or misleading elements.

Figure 9

The figure displays a diverse gallery of images, but the caption is misleading by excluding non-human subjects (a dog, a masked character) from its description of ‘human subjects,’ and the lack of sub-panel labels hinders specific referencing.

Figure 10

Figure 10 is a well-constructed image gallery that effectively supports its caption by showcasing diverse portrait generations with high aesthetic quality and fine-grained details. The visual content aligns perfectly with the claims of identity diversity and varied cultural settings.

Figure 11

Figure 11 effectively demonstrates the model’s text rendering capabilities across a diverse range of visual contexts, including murals, neon signs, product labels, and environmental signage. The generated text is legible and contextually appropriate, fully supporting the caption’s claim of showcasing typography in posters, signs, and graphic designs.

Figure 12

Figure 12 effectively demonstrates the model’s text rendering capabilities across diverse contexts including storefronts, signs, and labels. The visual quality is high, and the text is legible, supporting the caption’s claim of covering diverse visual contexts.

Action items.

- **[writing]** Figure 1: The caption contains a grammatical error and missing reference: ‘Section

provides more visualizations’ lacks a specific section number or name.

- **[writing]** Figure 1: The caption claims ‘1440 resolution’ but does not specify the dimension (e.g., 1440x1440 or 1440x2560), which is ambiguous for image generation tasks.
- **[science]** Figure 2: The caption states ‘marker area is proportional to model size’, but the plot contains no legend or scale to define this relationship, making the model size data unreadable.
- **[science]** Figure 2: The x-axis uses a logarithmic scale (1, 2, 5, 10...) but lacks explicit tick labels or grid lines to allow precise reading of inference times.
- **[writing]** Figure 2: The y-axis label ‘Score’ is ambiguous; the caption mentions two benchmarks (OneIG and GenEval) but does not specify which score is plotted or if they are combined.
- **[fatal]** Figure 3: The rendered image is a single pie chart that does not match the caption’s description of three distinct sub-figures (a, b, c) covering pre-training data, RL data, and caption length distribution.
- **[science]** Figure 3: The chart contains a logical contradiction where the central ‘Public’/‘Private’ circles are visually nested inside the ‘Real Data’ slice, yet the ‘Real Data’ slice is also labeled with a value (455.8M) that implies it is a separate category from the center.
- **[science]** Figure 3: The ‘Synthetic’ slice is labeled ‘1.844M’ but visually occupies a negligible sliver compared to the ‘Real Data’ (455.8M) and ‘Text-Synthetic’ (110M) slices, suggesting a potential unit error or mislabeling of the data.
- **[science]** Figure 4: The x-axis label ‘Training step (k)’ contradicts the caption’s claim of a ‘Caption-length ablation study’; the axis should represent caption length (e.g., word count) rather than training steps to validate the study’s premise.
- **[writing]** Figure 4: The caption is insufficient as it fails to define the specific caption lengths or word counts corresponding to the ‘Detailed’, ‘Mixed’, and ‘Brief’ categories shown in the legend.
- **[science]** Figure 6: The x-axis label ‘Training step (k)’ and the legend entry ‘GPT-OSS-20BA3B’ contradict the caption’s claim that this is a ‘Study of different language encoders’; the figure appears to show a training efficiency ablation (steps vs. score) rather than a comparison of encoder architectures.
- **[writing]** Figure 6: The x-axis tick labels are crowded and overlap (e.g., ‘104112120’), reducing legibility.
- **[science]** Figure 9: The image is a 4x3 grid of 12 distinct portraits, but the caption fails to label or number the sub-panels (e.g., a, b, c...), making it impossible to reference specific examples in the text.
- **[science]** Figure 9: The caption claims ‘diverse human subjects,’ yet the grid includes a dog (row 2, col 1) and a fantasy character with a neon mask (row 4, col 2), which contradicts the

stated scope of the figure.

Jargon Police

Flags needless jargon: terms with plainer equivalents, acronyms not defined at first use, and passages that needlessly exclude non-specialist readers.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_jargon_police.md

The manuscript relies heavily on domain-specific acronyms and jargon that are not defined at their first occurrence, creating barriers for non-specialist readers. In the Abstract, terms like ‘T2I’, ‘VAE’, and ‘RL’ are used immediately without expansion. Section 2.1 introduces ‘MMDiT’, ‘RoPE’, and ‘MoE’ without defining them. Section 2.5 uses ‘CFG’ without explanation. The Appendix introduces ‘LoRA’, ‘TTUR’, ‘IDA’, and ‘R1’ without definitions. Furthermore, benchmark acronyms like ‘NED’, ‘CVTG’, and ‘OneIG’ appear in tables and text without their full names or definitions in the immediate context. While these terms are standard in the specific sub-field of diffusion models, the paper’s claim of “rethinking training efficiency” suggests a broader audience interest, necessitating clearer definitions for all acronyms upon first use.

Action items.

- **[writing]** The manuscript relies heavily on domain-specific acronyms and jargon that are not defined at their first occurrence, creating barriers for non-specialist readers. In the Abstract, terms like ‘T2I’, ‘VAE’, and ‘RL’ are used immediately without expansion. Section 2.1 introduces ‘MMDiT’, ‘RoPE’, and ‘MoE’ without defining them. Section 2.5 uses ‘CFG’ without explanation. The Appendix introduces ‘LoRA’, ‘TTUR’, ‘IDA’, and ‘R1’ without definitions. Furthermore, benchmark acronyms like ‘NED’, ‘CVTG’,

Logical Consistency

Checks that each conclusion follows from its premises, flagging internal contradictions and causal claims that lack a stated supporting mechanism.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_logical_consistency.md

The manuscript presents a coherent argument for training efficiency, but several causal claims lack sufficient logical grounding regarding hardware normalization and generalization mechanisms.

First, the central efficiency claim in the Abstract and Introduction—that Lens requires “only about 19.3% of the training compute used by Z-Image”—is derived from a raw comparison of 192K A100 GPU hours versus 314K H800 GPU hours. While the text mentions TFLOPS (312 vs 989.5), it does not explicitly perform the normalization calculation in the text to justify the 19.3% figure. H800 GPUs offer significantly higher throughput for the specific tensor operations used in diffusion training compared to A100s. Without explicitly stating that the 19.3% figure is *after* normalizing for hardware performance (or clarifying that it is a raw hour comparison which

is misleading), the conclusion that the *algorithm* is 5x more efficient is not strictly supported by the premises provided. The logic assumes the reader will accept the raw hour ratio as a proxy for compute cost, which is a logical gap given the hardware disparity.

Second, the claim of generalization to resolutions up to 1440^2 and aspect ratios 1:2 to 2:1 “without explicit training” (Abstract, Section 3.3) is a strong causal assertion. The paper states the model was trained on 27 resolutions (max 1024^2) and 9 aspect ratios. While the use of RoPE (Rotary Positional Embeddings) is mentioned as a component, the text does not logically explain *how* the specific training distribution (up to 1024^2) enables robust generation at 1440^2 (a ~70% increase in pixel area). The mechanism of extrapolation is implied but not explicitly linked to the observed results in the text, leaving a gap between the premise (training on lower res) and the conclusion (high-res generalization).

Finally, the ablation study in Table 1 (Section 3.4) supports the claim that “prompt diversity is critical,” but the logical link to the specific “10 sample-aware + 1 global rubric” design is weak. The ablation varies the *presence of text prompts* in the RL dataset, not the rubric count or structure. Therefore, the conclusion that the specific rubric design is the driver of diversity is not directly supported by the provided evidence; the evidence only supports that *text-rich prompts* are necessary. The text conflates the dataset composition with the rubric mechanism.

Action items.

- **[science]** The 19.3% compute cost claim compares raw A100 vs H800 hours without normalizing for hardware throughput differences. The text asserts algorithmic efficiency but the premise (raw hours) does not logically support the conclusion (efficiency gain) without explicit TFLOPS-adjusted calculation.
- **[writing]** The claim of generalization to 1440^2 from 1024^2 training lacks a stated mechanism. The text asserts RoPE enables this but does not explain the causal link between the specific training distribution and the observed high-resolution extrapolation capability.
- **[writing]** Table 1 ablates text prompt presence, not rubric design. The conclusion that the specific ‘10+1 rubric’ design drives diversity is not logically supported by the provided evidence, which only shows text prompts are necessary.

Overreach

Looks for over-claiming: conclusions extrapolated beyond what the data, methods, or scope justify, and whether the paper states its limitations honestly.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_overreach.md

The manuscript makes several aggressive claims regarding training efficiency and generalization that slightly exceed the immediate evidence provided in the text.

First, the central efficiency claim in the Abstract and Introduction states that *Lens* requires “only about 19.3% of the training compute used by Z-Image.” This figure is derived by comparing 192K A100 GPU hours against 314K H800 GPU hours. However, the paper does not provide the

specific FLOP counts per hour for the A100 vs. H800 configurations used, nor does it normalize for the architectural differences (e.g., memory bandwidth, tensor core efficiency) that would affect the actual TFLOP-hours consumed. Without a normalized “compute cost” metric (e.g., total FLOPs), the 19.3% figure is an extrapolation based on raw time rather than a verified computational cost. The authors should either provide the normalized FLOP calculation or qualify the claim to refer specifically to “wall-clock time on the specified hardware” rather than “training compute.”

Second, the claim that the model “generalizes to... resolutions up to 1440^2 ” (Abstract, Section 1) suggests a robust capability at this specific resolution. While the paper mentions training on lower resolutions, it does not explicitly state whether 1440^2 was the upper bound of the *tested* generalization or a theoretical limit. If 1440^2 was the maximum resolution tested, the phrasing “up to” is acceptable but should be supported by a specific benchmark result or qualitative example at that exact resolution in the main text (currently, the teaser figure caption mentions 1440 resolution, but the specific performance metrics at this exact resolution are not detailed in the tables). If the model was not explicitly tested at 1440^2 , this is an over-extrapolation.

Finally, the assertion that strong language encoders “enable multilingual generalization from English-only training” (Section 1) is partially contradicted by the “Limitations” section, which admits the model “struggles with visual text rendering in non-English languages” and has lower accuracy. While the encoder may help the model *understand* multilingual prompts, the claim of “generalization” implies a level of output fidelity that the limitations section suggests is not fully achieved. The text should be refined to distinguish between “prompt understanding” (which the encoder aids) and “text rendering” (which remains a limitation).

Action items.

- **[science]** The claim that Lens achieves ‘19.3% of the training compute’ of Z-Image (Abstract, Intro) conflates GPU-hours with FLOPs without providing the specific FLOP counts or hardware efficiency factors for both models. The comparison of 192K A100 hours vs. 314K H800 hours is not directly additive without a normalized TFLOP-hour calculation, which is currently missing.
- **[writing]** The statement that the model ‘generalizes to aspect ratios 1:2 to 2:1 and resolutions up to 1440^2 ’ (Abstract, Intro) implies a hard boundary or guaranteed performance at these limits. The text should clarify if 1440^2 is the tested maximum or a theoretical extrapolation, and whether performance degrades significantly at the edges of this range compared to the training distribution.
- **[writing]** The claim that ‘Strong language encoders... enable multilingual generalization from English-only training’ (Intro) is a strong causal assertion. The paper admits in limitations that multilingual performance is ‘lower than English’ and struggles with non-English text rendering. The text should be tempered to reflect that the encoder aids *comprehension* of multilingual prompts but does not fully solve the *generation* of non-English text without specific training data.

Safety Ethics

Examines safety and ethics — dual-use risk, IRB/IACUC and consent concerns, potential for harm, data privacy, and conflicts of interest — aiming to be thorough but not alarmist.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_safety_ethics.md

The manuscript addresses safety and ethics primarily in the “Broader Impacts and Limitations” section (Section 7), acknowledging the risk of misuse for misleading content and the mitigation strategy of a “reasoner” module. However, the review identifies three critical gaps in the safety and ethics disclosure that require expansion before publication.

First, the claim that a reasoner “can identify and reject inappropriate user requests” (Section 1, Section 7) is a significant safety assertion that lacks empirical support or technical specification. The paper does not define the scope of “inappropriate” (e.g., hate speech, non-consensual deepfakes, violence), nor does it provide metrics on the reasoner’s false positive/negative rates. Without this, the claim is unverifiable and potentially misleading regarding the model’s actual safety posture. The authors must detail the safety taxonomy, the specific prompts used for the reasoner, and any evaluation of its effectiveness against adversarial inputs.

Second, the data provenance for the “Lens-800M” dataset is insufficiently transparent. The authors state the dataset includes “private” sources (Section 2.1) alongside public and synthetic data. The use of private data in foundational model training raises serious privacy and consent concerns. The manuscript must clarify the nature of this private data, the legal basis for its inclusion, and whether consent was obtained from the subjects depicted. Additionally, the reliance on GPT-4.1 for generating 800M dense captions introduces a layer of synthetic data whose own safety and bias properties are not discussed.

Third, the paper does not address the safety alignment of the large language models (GPT-4.1, GPT-5.5) used extensively for captioning, rubric generation, and reward modeling. If these upstream models contain biases or safety gaps, they are directly propagated into the Lens model’s training signal. A discussion on how these upstream risks are mitigated or monitored is necessary.

While the authors acknowledge limitations regarding multilingual performance and artifacts, the lack of concrete details on safety mechanisms and data ethics constitutes a significant omission for a foundational model paper.

Action items.

- **[writing]** The paper claims to use a ‘reasoner’ to reject inappropriate requests (Section 6, Section 7) but provides no details on the safety taxonomy, rejection rates, or failure modes. Explicitly describe the safety guardrails and their limitations to prevent misuse for generating harmful content.
- **[writing]** The training dataset (Lens-800M) includes ‘private’ sources and synthetic data generated by GPT-4.1. The manuscript lacks a clear statement on data provenance, consent for private data usage, and potential copyright or privacy violations inherent in the dataset composition.

- **[writing]** The paper relies on GPT-4.1 and GPT-5.5 for captioning, rubric generation, and reward modeling. There is no discussion of the safety alignment of these underlying models or the risk of propagating their biases into the Lens model’s generation capabilities.

Scientific Evidence

Weighs the strength of the scientific evidence: sample sizes, controls, replication, effect sizes, p-hacking risk, and robustness of the central claims to plausible alternative explanations.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_scientific_evidence.md

The manuscript presents a compelling efficiency narrative, but the scientific evidence supporting the central claims of “training efficiency” and “SOTA performance” requires stronger statistical grounding and hardware normalization.

First, the primary claim of achieving “19.3% of the training compute” (Abstract, Section 1) relies on a direct comparison of 192K A100 GPU hours against 314K H800 GPU hours. This comparison is methodologically flawed as it ignores the architectural differences between A100 and H800 GPUs. While the authors provide raw TFLOPS figures (312 vs 989.5), they do not integrate these into a unified compute metric (e.g., total FLOPs or normalized GPU-hours). Without normalizing for the specific hardware capabilities, the 19.3% figure is an apples-to-oranges comparison that overstates the efficiency gain. The evidence for this claim is currently insufficient to support the magnitude of the assertion.

Second, the benchmark results, particularly the GenEval score of 0.930 (Table 1), lack necessary statistical rigor. The paper reports single-point estimates without standard deviations, confidence intervals, or the number of random seeds used for evaluation. In generative modeling, performance can vary significantly based on sampling seeds and prompt variations. The absence of variance metrics makes it impossible to determine if the reported improvements over competitors (e.g., LongCat-Image at 0.870) are statistically significant or within the margin of error. Furthermore, the heavy reliance on prompt rewriting (Section 4) introduces a confounding variable; without reporting the variance across different prompt rewriting seeds or strategies, the robustness of the 0.930 score is questionable.

Finally, the ablation study on RL dataset diversity (Table 2) presents a marginal improvement (0.916 to 0.930) as a critical finding. The text asserts that “full diversity is critical” based on this ~1.4% absolute gain. However, without a statistical significance test (e.g., a t-test) or error bars, it is unclear if this difference is distinguishable from noise. Given the relatively small size of the RL dataset (8K prompts), the statistical power to detect such small effects is low. The authors should provide statistical validation to support the claim that the full dataset is strictly necessary versus a smaller subset.

In summary, while the engineering contributions appear sound, the scientific evidence regarding efficiency claims and performance robustness needs to be strengthened with normalized compute metrics and rigorous statistical reporting.

Action items.

- **[science]** The claim that Lens achieves 19.3% of Z-Image’s compute cost (Abstract, Intro) compares 192K A100 hours to 314K H800 hours. This conflates hardware generations (A100 vs H800) and does not normalize for FLOPS or memory bandwidth. The paper cites raw TFLOPS (312 vs 989.5) but fails to provide a unified compute metric (e.g., total FLOPs or normalized GPU-hours) to substantiate the ‘19.3%’ figure. Re-calculate efficiency using a hardware-agnostic metric.
- **[science]** The GenEval score of 0.930 (Table 1) is exceptionally high compared to prior SOTA (e.g., LongCat-Image 0.870). The paper does not report standard deviation, confidence intervals, or the number of seeds used for evaluation. Given the sensitivity of GenEval to prompt rewriting (Section 4, ‘Prompt for GenEval Benchmark’), the lack of statistical variance reporting makes it impossible to assess if the improvement is robust or an artifact of specific prompt engineering.
- **[science]** The ablation study on RL dataset diversity (Table 2) shows GenEval scores of 0.916 (1/4 set) vs 0.930 (Full set). The absolute gain is small (1.4%), yet the text claims ‘full diversity is critical.’ The paper lacks a statistical significance test (e.g., t-test) to determine if this difference is meaningful or within the noise of the evaluation metric, especially given the small sample size of the RL set (8K prompts).

Statistical Analysis

Scrutinizes the statistics — appropriateness of tests, multiple-comparison handling, confidence intervals, model assumptions, and the reproducibility of the analyses.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_statistical_analysis.md

The manuscript presents a compelling narrative for training efficiency in text-to-image models, but the statistical rigor supporting the quantitative claims requires strengthening.

First, the central claim of “19.3% of the training compute” (Abstract, Section 1) relies on a direct comparison of GPU hours between A100 (Lens) and H800 (Z-Image) hardware. This is a methodological flaw in statistical comparison; H800s offer significantly higher FLOPs per hour than A100s. Without normalizing these figures to a common compute unit (e.g., total FLOPs or H100-equivalent hours), the efficiency claim is not statistically sound. The authors must provide a normalized compute metric to validate the magnitude of the efficiency gain.

Second, the benchmark results presented in Table 1 (Main Results) and the Appendix tables (e.g., GenEval, OneIG) report single-point estimates (e.g., 0.930 for GenEval) without any measure of uncertainty. Diffusion model generation is inherently stochastic, and benchmark scores can vary based on random seeds and sampling schedules. The absence of confidence intervals, standard deviations, or results averaged over multiple independent runs makes it impossible to determine if the observed differences between Lens and baselines (e.g., Qwen-Image, Z-Image) are statistically significant or within the margin of error.

Finally, the ablation study on RL dataset diversity (Table 2) presents a monotonic improvement

trend (0.916 -> 0.920 -> 0.930) based on only three data points. While the trend is logical, the lack of variance reporting or significance testing (e.g., t-tests) leaves the robustness of this finding unverified. Given the high cost of RL training, confirming that the full dataset provides a statistically significant improvement over the 1/2 subset is crucial.

The authors should re-run key experiments with multiple seeds to report variance, normalize the compute efficiency metrics, and include statistical significance tests for all comparative claims.

Action items.

- **[science]** The paper claims Lens achieves ‘19.3% of the training compute’ of Z-Image (Abstract, Intro) based on 192K A100 hours vs. 314K H800 hours. This comparison conflates hardware generations (A100 vs. H800) and lacks a normalized FLOPs calculation. Please provide a normalized compute estimate (e.g., total FLOPs or H100-equivalent hours) to substantiate the efficiency claim.
- **[science]** Benchmark results in Table 1 and Appendix tables report single-point scores (e.g., GenEval 0.930) without confidence intervals, standard deviations, or p-values. Given the stochastic nature of diffusion sampling and the finite size of benchmarks (e.g., 553 prompts for GenEval), statistical significance testing or variance reporting is required to validate performance differences against baselines.
- **[science]** The RL ablation in Table 2 (RL Training Set diversity) shows performance gains with full data but lacks statistical validation. With only three data points (1/4, 1/2, Full), the trend is suggestive but not statistically robust. Please report variance across multiple seeds or apply a significance test to confirm the improvement is not due to random fluctuation.

Writing Quality

Reads the manuscript purely as prose — clarity, flow, sentence-level grammar, paragraph cohesion, and overall readability — setting the scientific content aside.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_writing_quality.md

The manuscript presents a compelling narrative regarding training efficiency, with generally clear and professional prose. The abstract effectively summarizes the core contributions, and the logical flow from problem statement to methodology and results is sound. However, several instances of ambiguous phrasing and inconsistent terminology hinder immediate comprehension for a general reader.

Specifically, the introduction’s comparison of compute costs uses “314K H800 GPU hours” without clarifying the unit of measurement (e.g., total hours vs. cluster-hours), which is critical for validating the claimed 19.3% efficiency gain. In Section 2.1, the data cleaning pipeline lists acronyms like “EVA” and “Aesthetic Predictor” without defining them or providing immediate context, forcing the reader to search citations for basic definitions. Additionally, Figure 1’s caption refers to “1440 resolution,” whereas the text consistently uses “1440²” or “1440x1440”;

this inconsistency should be resolved to prevent confusion between linear dimensions and pixel area.

The definition of the “CVTG” benchmark in Section 4 is also delayed, appearing only in the table caption rather than the main text where the benchmark is first discussed. Finally, the description of the 27 training buckets in Section 2.3 is slightly dense; a minor rephrasing would improve readability. Addressing these specific clarity issues will significantly enhance the paper’s accessibility without altering its scientific content.

Action items.

- **[writing]** In Section 1 (Introduction), the phrase ‘314K H800 GPU hours’ is ambiguous. Clarify if this refers to total compute hours or a specific cluster size over time to ensure the efficiency comparison with Lens is mathematically precise.
- **[writing]** Section 2.1 (Pre-training Data) lists ‘NSFW (EVA...)’ and ‘aesthetic (Aesthetic Predictor...)’ without defining the acronyms or model versions upon first mention. Define ‘EVA’ and ensure the citation for the aesthetic predictor is clear.
- **[writing]** The caption for Figure 1 (teaser) states ‘1440 resolution’ but the text and other sections specify ‘1440²’ (pixels). Standardize this to ‘1440²’ or ‘1440x1440’ to avoid confusion between linear dimension and area.
- **[writing]** In Section 2.3 (Pre-training), the phrase ‘3 base areas... x 9 aspect ratios (27 buckets)’ is slightly dense. Consider rephrasing to ‘3 base resolutions combined with 9 aspect ratios, forming 27 buckets’ for better flow.
- **[writing]** Section 4 (Comparison) introduces ‘CVTG’ without a full expansion or definition in the main text, relying on the caption or external knowledge. Define ‘Complex Visual Text Generation (CVTG)’ upon first use in the main body.